Pharmacology of Anemias

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Following completion of the learning materials, students should be able to:

1. Identify the goals of therapy for the microcytic, normocytic, and megaloblastic anemias covered in this presentation.
2. Relate the pharmacokinetic and biochemical properties of iron, folic acid and vitamin B12 supplements to their effects on laboratory values in the management of specific anemias.
3. Correlate the pharmacokinetic properties and mechanisms of the erythropoiesis-stimulating agents (ESAs) to their effects on laboratory values in the management of anemia of chronic kidney disease and chemotherapy-induced anemia in selected cancer patients.
4. Relate the pharmacokinetic properties and mechanisms of hydroxyurea to its effects in the management of sickle cell anemia.
5. Explain the side effects, adverse effects, cautions, contraindications, and drug interactions of the various products in relation to their mechanisms and effects.

Preparation Materials (links are in the CPG and on the next slide)

Required

- Practice Questions

Optional materials for acquiring new knowledge:

- Video Lecture | Dr. Goldstein's Notes handout | Guided Reading Questions
- ScholarRx Bricks and Textbook resources with links are listed on the next slide.

SUGGESTIONS:

- *Use the resources that work best for you.*
- *You do not need to study all of them.*
- *Work through the GUIDED READING QUESTIONS with pen/pencil and paper.*

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

- *Practice questions: Active recall boosts long-term retention.*

Links to resources

ScholarRx Brick: Name of Discipline/ Organ System: Hematology

- Iron deficiency anemia: https://exchange.scholarrx.com/brick/iron-deficiency-anemia
- Megaloblastic anemia: https://exchange.scholarrx.com/brick/megaloblastic-anemia
- Sickle cell anemia: https://exchange.scholarrx.com/brick/sickle-cell-disease
- Anemia Review: https://exchange.scholarrx.com/brick/anemias-putting-it-all-together

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 33: Agents Used in Cytopenias; Hematopoietic Growth Factors (relevant sections)

https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3382&sectionid=281752393

LWW Health Library: Premium Basic Sciences; Lippincott's Illustrated Reviews: Pharmacology: Chapter 44: Drugs for Anemia https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=253330632&bookid=3222

Access Medicine Katzung's Pharmacology: Examination & Board Review, 13e, 2021; Chapter 33: Agents Used for Cytopenias; Hematopoietic Growth Factors https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3461&sectionid=285595479#1207302166

ANEMIA: A significant deficit in the mass of circulating oxygen-carrying red blood cells.

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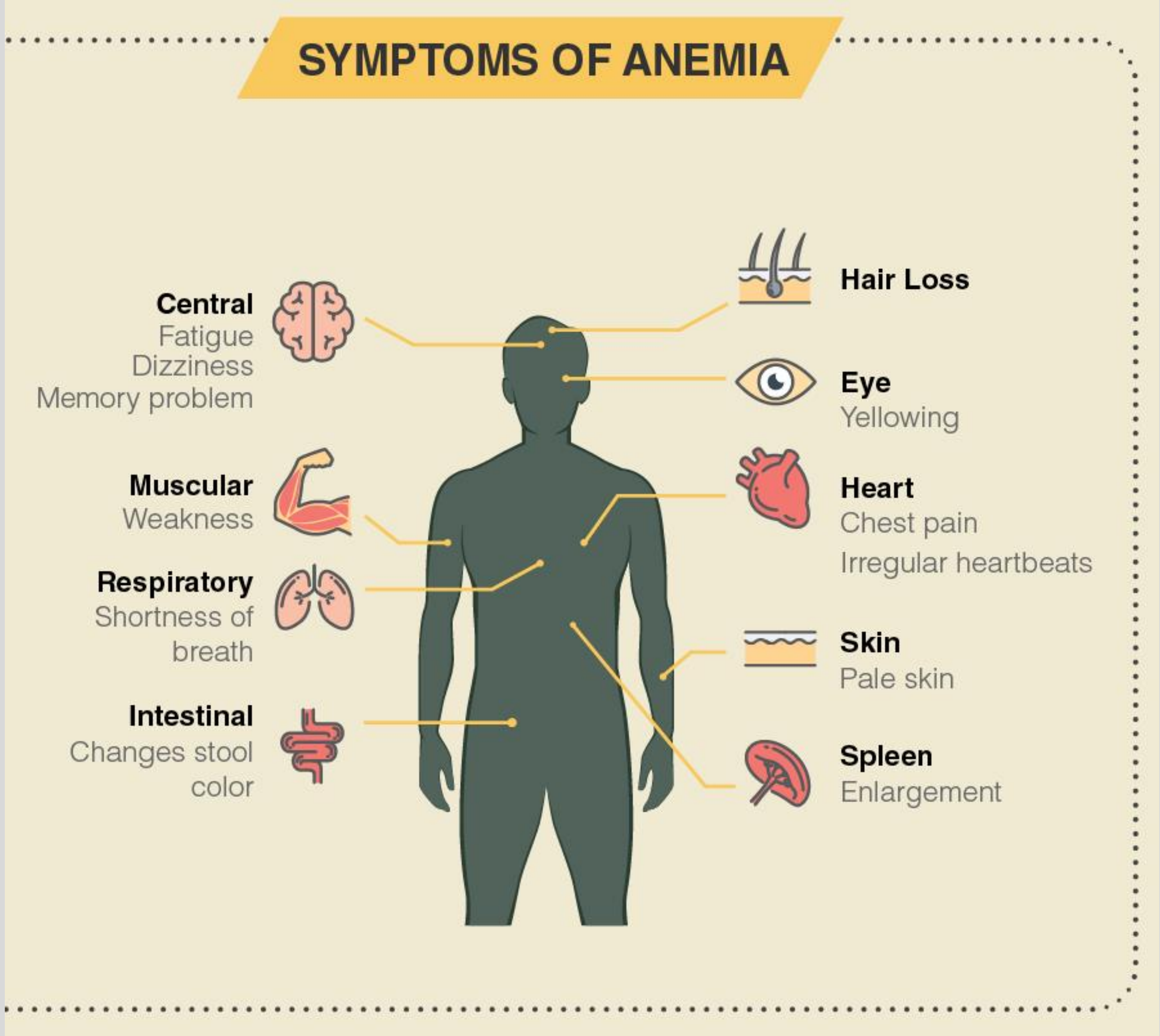
Features

- Hemoglobin level or hematocrit is more than two standard deviations below the lower limit of normal for age and gender.
- Erythropoietin levels are increased in direct proportion to the severity of anemia.
- Some underproduction anemias are associated with ineffective erythropoiesis.
- ***Management is directed at the cause.***

Symptoms of anemia

- fatigue, pale skin, weakness, dizziness, and shortness of breath
- headaches, heart palpitations, cold hands/feet, and brittle nails.



https://www.lalpathlabs.com/blog/signs-symptoms-causes-and-treatment-of-anemia/

Key points

Anemias are classified as:

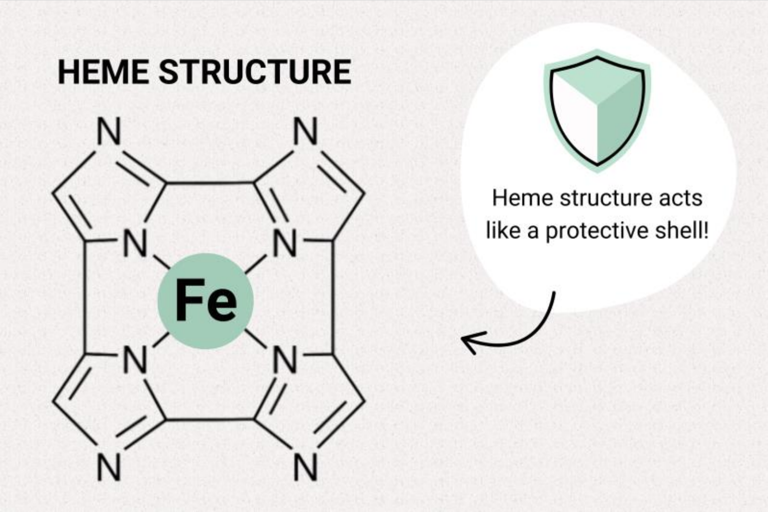
- Microcytic hypochromic anemia, caused by iron deficiency, is the most common type of anemia.
- Normocytic, normochromic anemia is due to chronic inflammation, decreased erythropoietin production (chronic kidney disease), acute blood loss, hemolysis, and others.
- Megaloblastic anemia is a subset of the macrocytic, normochromic anemia caused by a deficiency of folic acid or vitamin B_{12}, cofactors required for the normal maturation of red blood cells.
- Pernicious anemia, the most common type of vitamin B_{12} deficiency anemia, is caused by an autoimmune attack on gastric parietal cells and intrinsic factor, a protein required for efficient absorption of dietary vitamin B_{12}, or by surgical removal of that part of the stomach that secretes intrinsic factor.
- Erythropoietin, which is produced by the specialized cells of the kidney, activates receptors on erythroid progenitors in the bone marrow, thereby stimulating the production of red cells and increasing their release from the bone marrow.
- Recombinant erythropoiesis stimulating agents (ESAs) may be used for the treatment of symptomatic anemia not responding to therapy for the underlying cause of the disease. The ESAs in use in the U.S. are epoetin alfa, darbepoetin alfa, and methoxy polyethylene glycol-epoetin beta.
- The most common complications of ESA therapy are hypertension and thrombosis, leading to myocardial infarction and stroke. Serum hemoglobin in patients treated with an ESA should not exceed 10-11 g/dL because higher hemoglobin levels are associated with increased mortality and cardiovascular adverse events.

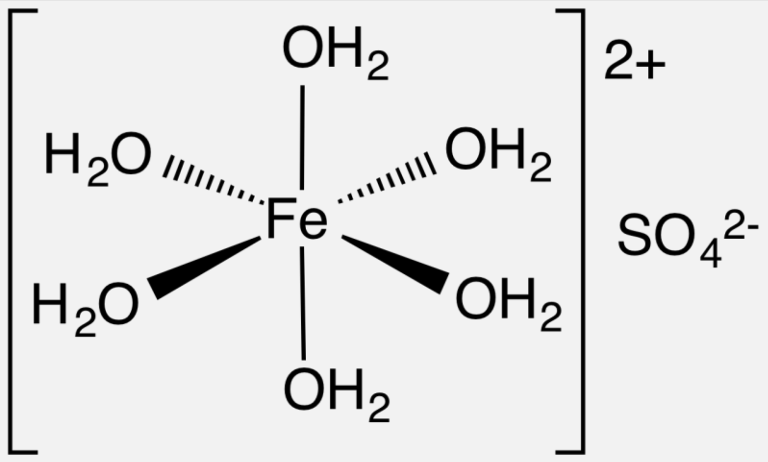
Key points

- Sickle cell disease is a genetic cause of hemolytic anemia due to increased erythrocyte destruction.
- Patients with sickle cell disease are homozygous for the abnormal β-hemoglobin S (*HbS*) allele. Deoxygenated HbS chains form polymeric structures that dramatically change erythrocyte shape (sickle), reduce deformability, and elicit membrane permeability changes that promote hemoglobin polymerization, eventually leading to veno-occlusive events causing pain and organ ischemia.
- The goal of therapy with disease modifying therapies for sickle cell anemia is to reduce the frequency of painful veno-occlusive crises and the need for blood transfusions.
- Hydroxyurea is the first-line medication for treatment of sickle cell anemia. It is an antimetabolite that selectively inhibits ribonucleoside reductase, preventing the conversion of ribonucleotides to deoxyribonucleotides, halting the cell cycle at the G1 to S phase.
- In sickle cell anemia, hydroxyurea increases red blood cell (RBC) hemoglobin F levels, RBC water content, deformability of sickled cells, and alters adhesion of RBCs to endothelium. It is generally well tolerated but can cause myelosuppression, GI disturbance, and dermatologic reactions.
- Crizanlizumab or L-glutamine may be added to hydroxyurea therapy if pain episodes are still present after 6 months or more. Crizanlizumab is an anti-P-selectin antibody that reduces interactions between endothelial cells and circulating blood cells. L-glutamine may improve the NADH/NAD+ redox balance, reducing oxidative damage, decreasing sickling, and improving membrane integrity of RBCs.

Iron Deficiency Anemia

Iron Supplements





Background: Physiologic Iron Regulation and Dietary Sources

Regulation:

Iron balance is achieved by changing intestinal absorption and storage of iron in response to the body's needs. Less iron is absorbed when iron stores are adequate. More iron is absorbed when iron stores are low.

- ~10% of an oral dose is absorbed when iron stores are normal.
- 20-30% of an oral dose is absorbed when iron stores are inadequate.
- The relative percentage of absorbed iron decreases with increasing dose.

In an iron-deficient individual, about 50-100 mg of iron can be incorporated into hemoglobin daily

Dietary sources of Iron:

Animal-based (Fe^{2+}): Heme iron

Plant-based (Fe^{3+}): Leafy greens, legumes, nuts/seeds, dried fruits, fortified foods

Gastric acid allows Fe^{3+} dissociation from food proteins $\rightarrow$ soluble Fe^{3+}.

At pH>5 $\rightarrow$ Fe^{3+} forms highly insoluble $Fe(OH)_3$.

Vitamin C is a strong reducing agent (antioxidant): Food rich in vitamin C (citrus fruits, bell peppers, broccoli) increase absorption of non-heme iron by reducing $Fe^{3+} \rightarrow Fe^{2+}$.

Polyphenols / tannins and calcium form insoluble complexes with iron. Drinking tea, coffee or calcium-rich milk during meals can reduce iron absorption.

Iron Absorption, Transport, Metabolism, and Storage

1. Iron is absorbed via divalent metal transporter 1 (DMT1) and heme carrier protein 1 (HCP1). (Fe3+ is reduced by duodenal cytochrome b, DB.)

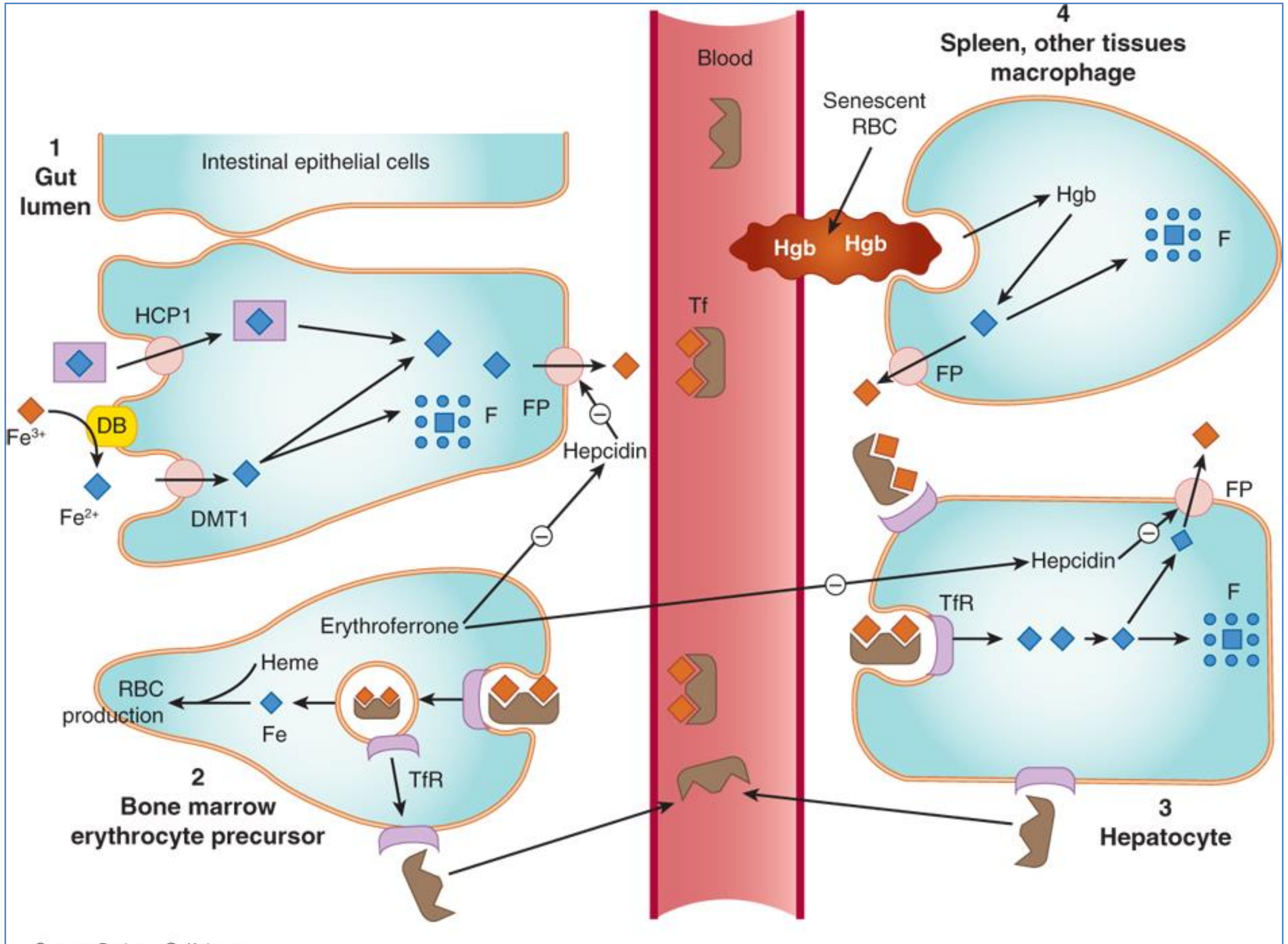
- Iron is stored as ferritin (F) and exported via ferroportin (FP) and
- Transported to bone marrow or hepatocytes as transferrin (Tf)-Fe3+ iron complex
- Internalized by binding of Tf-iron complex to transferrin receptors (TfR)

2. Iron in erythrocyte precursors is used for heme synthesis.

3. Iron in hepatocytes is stored as ferritin.

4. Iron is reclaimed from senescent RBCs by macrophages that phagocytize them and exported or stored as ferritin.

Hepcidin is a peptide hormone synthesized in hepatocytes. It regulates iron absorption, recycling, and release by binding ferroportin, which is internalized→ ↓ serum iron levels.



Katzung Basic & Clinical Pharmacology, 14e, 2021; Figure 33-1

Ferrous iron (Fe^{2+}), blue diamonds, squares; ferric iron (Fe^{3+}), red; DB, duodenal cytochrome B; F, ferritin

Iron Replacement Products

Goals of therapy: To fully replace iron balance, identify and treat the underlying cause, and prevent recurrence

Oral supplementation

- Effective
- Safe
- Readily available
- Inexpensive

Take on empty stomach:

- Coffee, tea, dairy, fiber, many foods can reduce absorption.

IV administration preferred in the following:

- Oral iron is ineffective or poorly tolerated
- Severe or ongoing blood loss
- Gastric bypass or resection→ ↓ gastric acid
- Malabsorption syndromes
- Chronic diseases (inflammation)
- High iron demand such as hemodialysis, chemotherapy, or erythropoietin use

Monitor lab values:

- Serum ferritin directly measures iron stores. Low level is the most sensitive and specific test.
- Other critical markers: low hemoglobin/hematocrit, low MCV, low serum iron, high total iron binding capacity

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Product	Amount of elemental iron	Comments
✳Ferrous sulfate 325 mg tablet 220 mg/5 mL solution	65 mg 44 mg/5mL	• Most common formulation • Generally tolerable • 2-4 x daily for anemia tx
✳Iron dextran, low molecular weight◊ Colloidal solution of ferric oxychloride complexed with dextran	50 mg/mL	• Test dose prior to first dose◊ • IV infusion: One large dose or multiple doses • Anaphylaxis potential • Delayed hypersensitivity reactions, especially in patients with history of allergies or rheumatoid arthritis • Processed by macrophages

Several more oral iron products are available. They differ in their iron content and tolerability.

Several more IV iron products are available. Test doses are not required. **All are equally efficacious.**

◊Fatal hypersensitivity / anaphylactoid reactions occurred with high molecular weight iron dextran (HMW ID) formulations. This product has been withdrawn from the market.

Adverse Effects and Drug Interactions of Iron Replacement Products

Oral supplementation:

Local GI irritation

- Abdominal pain
- Nausea
- Constipation
- Diarrhea
- Dark stools (harmless)

IV administration:

- Fatal anaphylactoid (IgE-mediated) reaction
    - Iron dextran, high molecular weight mainly
    - Prior reaction to HMW ID is the strongest predictor.

Infusion related:

- Flushing, headache, dizziness, metallic taste, nausea, myalgias, arthralgias, transient hypotension
- May exacerbate asthma or inflammatory arthritis (premedication may be indicated)

Drug interactions:

Iron can reduce absorption of levothyroxine, quinolone antibiotics, levodopa, phosphate supplements, many others.

Iron absorption can be reduced by antacids, bisphosphonates, tetracyclines.

Doses should be separated. Timeframe recommendations vary by drug combinations.

Check your knowledge

What are the goals of iron supplementation therapy?

What is the first-line treatment of iron deficiency anemia?

What laboratory value best reflects total body iron stores?

Which form of iron is absorbed by enterocytes.

Why should oral iron supplements be taken on an empty stomach?

Check your knowledge

How does the body regulate the amount of iron that is absorbed?

What are some important drug interactions?

When is IV iron infusion preferred?

What vitamin enhances oral absorption of iron?

Why is gastric acid important for absorption of iron from plant-based food?

Folate and Vitamin B12 Deficiencies

Brief background

Folic acid and vitamin B12 are coenzymes for various metabolic functions, including the purine and thymidylate synthesis for the synthesis of DNA and proteins necessary for cell replication and erythropoiesis and in fat and carbohydrate metabolism.

Deficiency of either folate or B12 decreases the synthesis of methionine and s-adenosyl-methionine (SAM), which interferes with DNA synthesis, protein synthesis, several methylation reactions, and synthesis of polyamines and results in cellular maturation abnormalities.

- Nuclear maturation is delayed → fewer RBCs → anemia
- RNA and protein synthesis continue → cytoplasm grows → megaloblasts → megaloblastic anemia.

Deficiency of vitamin B12 (but not folate) can lead to impaired formation of the myelin sheath, resulting in neurological symptoms:

Paresthesia in peripheral nerves and weakness; progresses to spasticity, ataxia, neuropsychiatric changes (may be irreversible).

Folic Acid Deficiency

- Folic Acid Supplementation

Folic Acid Metabolism and Deficiency

Folic acid is a water-soluble vitamin.

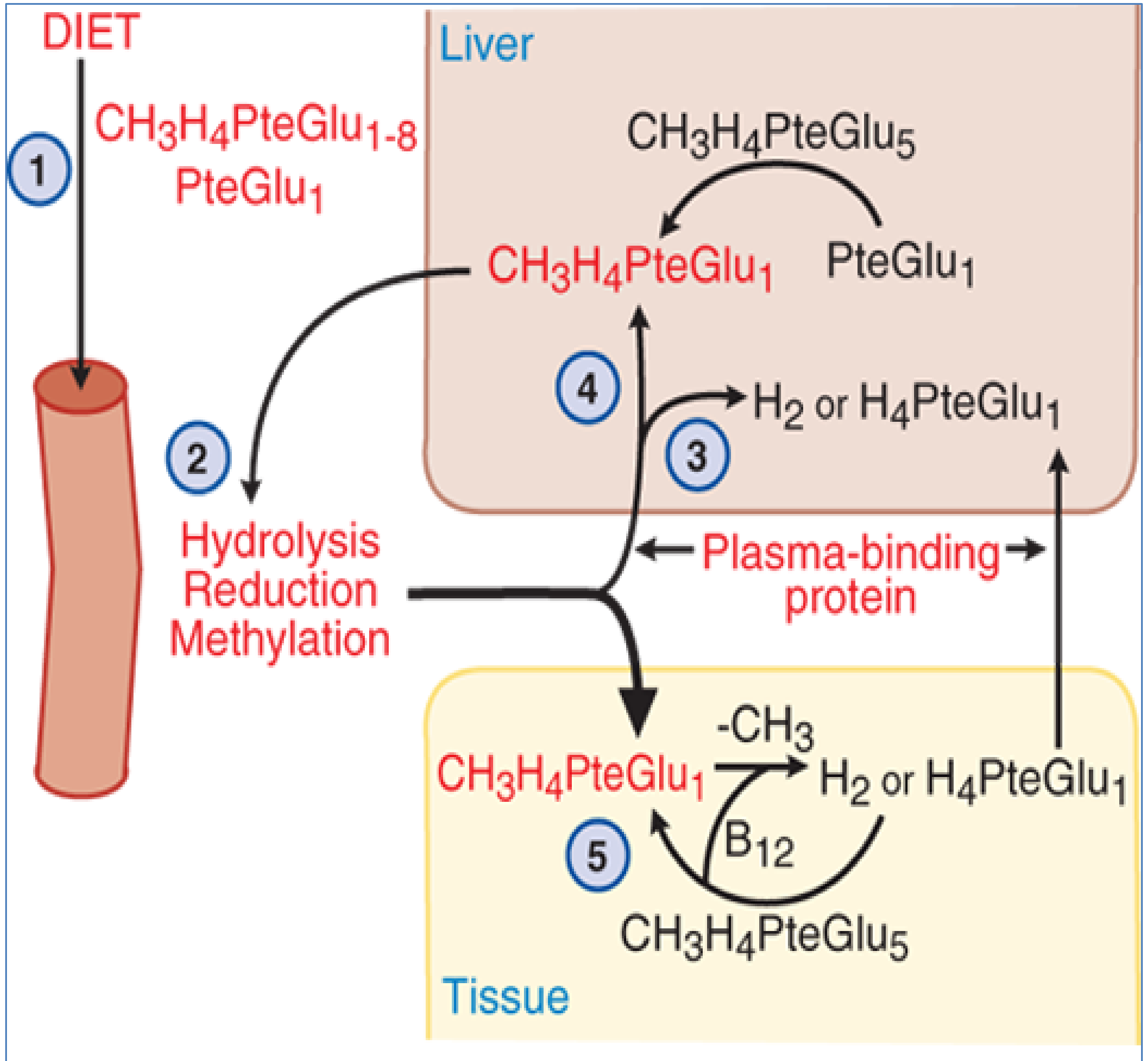
Dietary folate is present as polyglutamated folate. These polyglutamates are hydrolyzed to monoglutamate forms by brush-border glutamate carboxypeptidase.

Folic acid is then reduced by dihydrofolate reductase (DHFR) in the mucosa of the duodenum and proximal jejunum to tetrahydrofolate (THF), subsequently methylated in the liver, secreted into bile, and undergoes enterohepatic circulation (the liver is major regulator of folate).

Folate is stored as polyglutamates in hepatocytes and other tissues.

Folate deficiency causes a shortage of THF needed for DNA synthesis and stalls cell division in RBC precursors. Folate depletion takes several months to occur.

Folate deficiency does not cause the characteristic neurologic syndrome seen in vitamin B12 deficiency.



PteGlu = pteroyl-L-glutamic acid = folic acid

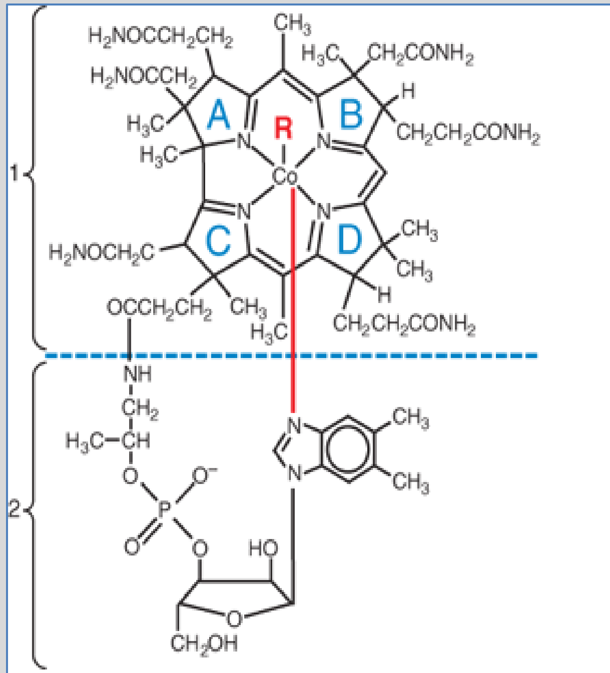
Goodman & Gilman's 2023: Figure 45-9

Folic Acid (vitamin B9)

PK	Oral; 85-100% bioavailability with and without food | IM, subQ, IV (rarely necessary) urine excretion Dose: 1 mg to 5 mg daily (0.4 mg maintenance dose)
Goal	For the treatment of megaloblastic anemia, the goal is to reverse folate deficiency, stimulate production of healthy RBCs, and achieve complete hematological remission.
Effects	Reticulocyte count increases in 3 to 4 days Hgb and Hct improve in ~1-2 weeks; normalize in 4 to 8 weeks
Caution	***Avoid folic acid monotherapy administration in patients with vitamin B12 deficiency.*** ➢ Folic acid monotherapy is appropriate *only* when laboratory testing confirms low RBC and serum folate levels and normal B12 levels. ***Why?*** 1. Folic acid monotherapy unmasks vitamin B12 deficiency by correcting anemia but allowing neurological damage to progress unnoticed. 2. Worsening neurological complications of vitamin B12 deficiency may be irreversible. Folate alone does not prevent or treat neurologic damage from B12 deficiency.
	Drugs that reduce folate absorption or metabolism: A) Phenytoin, sulfasalazine, alcohol M) DHFR inhibitors: methotrexate, trimethoprim, pyrimethamine

Vitamin B12 Deficiency

- ✳Cyanocobalamin
- ✳Hydroxocobalamin



Vitamin B_{12} Congeners

Permissive Name	*R Group*
Cyanocobalamin (Vitamin B_{12})	–CN
Hydroxocobalamin	–OH
Methylcobalamin	$–CH_3$
5'-Deoxyadenosylcobalamin	–5'-Deoxyadenosyl

Source: Laurence L. Brunton, Randa Hilal-Dandan, Björn C. Knollmann: Goodman & Gilman's: The Pharmacological Basis of Therapeutics, Thirteenth Edition:

Goodman & Gilman, 2018, Figure 41-5

Risk factors for B12 deficiency:
vegan or strict vegetarian diet
gastric or bariatric surgery

Vitamin B12 is stored in the liver. If vitamin B12 cannot be absorbed, as in people with dietary insufficiency, malabsorption, or lack of intrinsic factor, hepatic stores are depleted, resulting in vitamin B12 deficiency. (It usually takes 3 to 5 years to deplete the stores.)

Goals of vitamin B12 therapy in megaloblastic anemia is to rapidly correct the deficiency, restore hematologic parameters, and prevent or reverse neurologic damage.

- B12 (methylcobalamin) is a cofactor of methionine synthase in thymidylate synthesis and various other biochemical interactions.

“Methyl-folate trap” leads to megaloblastic anemia:

- MTHF is not converted to THF when B12 is absent, which leads to functional folate deficiency even if dietary folate intake is adequate.

Impaired methylmalonyl-CoA mutase function leads to disruption of myelin sheath production → neurologic damage

- B12 deficiency causes dysfunction of the enzyme. Methylmalonyl-CoA and propionyl-CoA accumulate resulting in production of abnormal fatty acids that are incorporated into neuronal lipids and disrupt the structure and function of the myelin sheath.
- Paresthesia in peripheral nerves and weakness; progresses to spasticity, ataxia, neuropsychiatric changes (may be irreversible).

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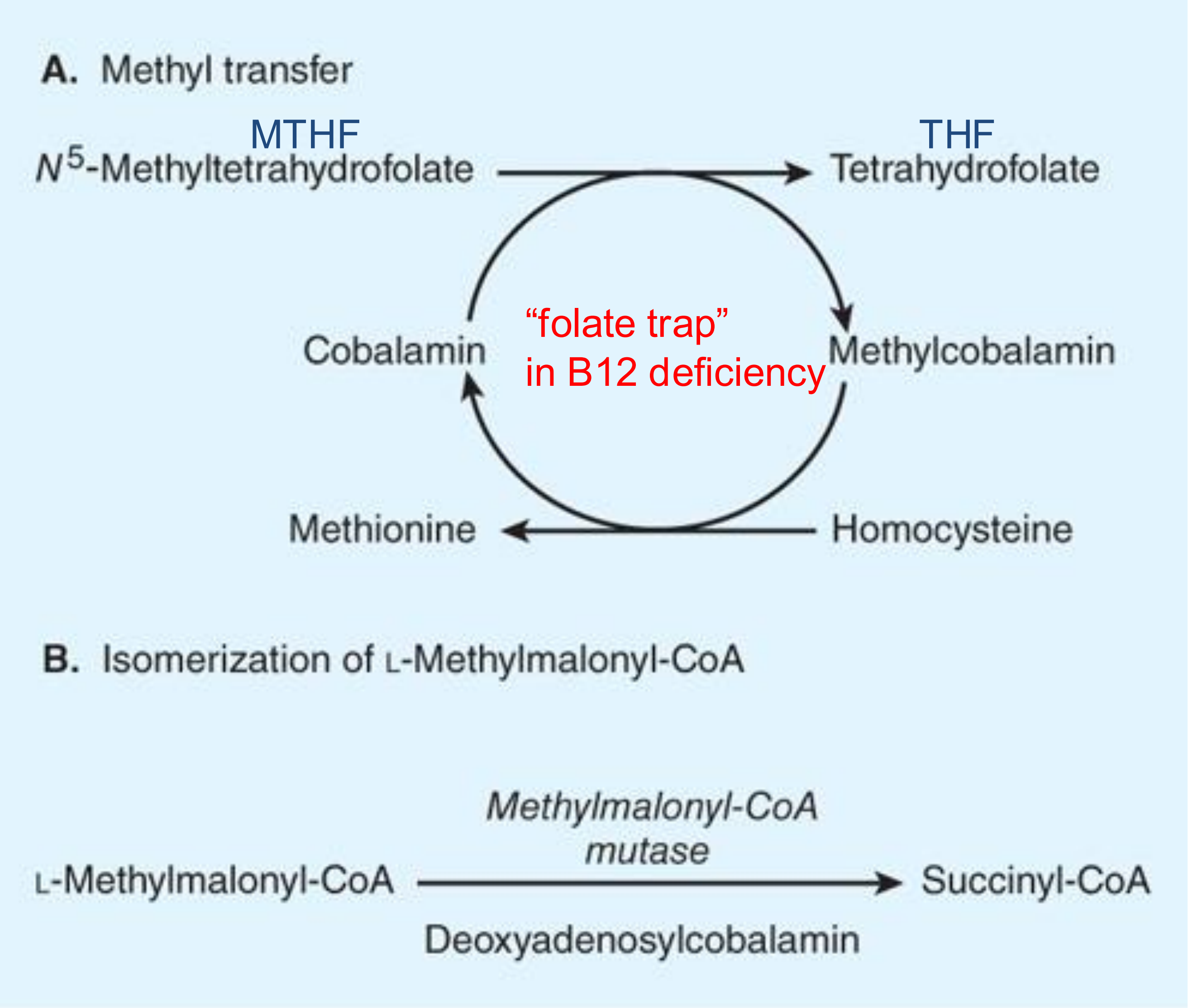
1. Methylcobalamin is an intermediate in the transfer of a methyl group from N^5-methyltetrahydrofolate to homocysteine, forming methionine.
2. Isomerization of methylmalonyl-CoA to succinyl-CoA by methylmalonyl-CoA mutase



Source: Todd W. Vanderah:
Basic & Clinical Pharmacology, Sixteenth Edition

Katzung, 2024, Figure 33-2
Enzymatic reactions that use vitamin B_{12}

Pharmacokinetics

	Cyanocobalamin	Hydroxocobalamin
A	Oral tablet, sublingual liquid and tablet, nasal solution, IM injection, deep subQ injection Dosing: 1000 mcg daily, 1-2 weeks then monthly maintenance	IM for the treatment of B12 deficiency IV or IO for treatment of cyanide poisoning 1000 mcg IM maintenance, anemia every 2-3 months Cyanide poisoning 5 gm x2 IV / IO infusion
	Oral absorption is variable from the terminal ileum. Intrinsic factor and calcium are required to transfer the compound across the intestinal mucosa. Cyanocobalamin oral bioavailability: 1.2% in patients with pernicious anemia → IM administration is necessary. Nasal 6.1% (relative to IM) → useful for maintenance in patients with pernicious anemia or malabsorption	
D	Transcobalamins: Specialized glycoproteins that transport vitamin B12 to the various cells of the body Hydroxocobalamin is more highly protein bound and remains in circulation longer → longer duration.	
M	Cyanocobalamin is converted in tissues to active coenzymes, methylcobalamin and deoxyadenosylcobalamin. The cyanide is metabolized in the liver to nontoxic thiocyanate.	Hydroxocobalamin is also converted to methylcobalamin and deoxyadenosylcobalamin.
E	urine/feces 50% to 98% unchanged Thiocyanate is excreted in urine.	Urine, 50-60% within 72 hours
	Storage: 90% (1 to 10mg) is stored in the liver as active coenzyme. It is also stored in bone marrow, kidneys, and adrenal glands.	

Check your knowledge

What vitamin deficiencies lead to megaloblastic anemia?

How do the mechanisms of anemia by the two vitamins differ?

Which vitamin deficiency leads to neurological symptoms and why?

Check your knowledge

How does absorption of folic acid and vitamin B12 differ?

Erythropoiesis-stimulating agents (EPAs)

✳Epoetin alfa: recombinant human EPO

Epoetin alfa biosimilars

Darbepoetin alfa: glycosylated EPO → longer duration

Methoxy polyethylene glycol-epoetin beta (modified EPO structure) → much longer duration

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ESAs: Pharmacokinetics

	Epoetin alfa		Darbepoetin alfa
A	IV, SubQ (slow absorption)		SubQ; slow absorption
Bioavailability	Adults: 36% Neonates: 42%		Patients with chronic kidney disease (CKD): Adults: 30-50% Children: 32-70%
	Large molecules diffuse into lymphatics – slow and capacity limited. Proteases, peptidases, macrophages in the interstitial space degrade the molecule.		
Administration frequency	3 times weekly (not significantly removed by dialysis)		once every 1-2 weeks (not significantly removed by dialysis)
D	Erythropoietin concentrates in liver, bone marrow, and kidneys		
$t_{1/2\,elimination}$	Adults, subQ: 16-67 h	CKD: IV Adults: 4-13 h Children: 5-24 h	35-139 hours depends on kidney function and individual metabolic rates
Onset	Hgb/Hct increase 2-6 weeks after initiating Tx		

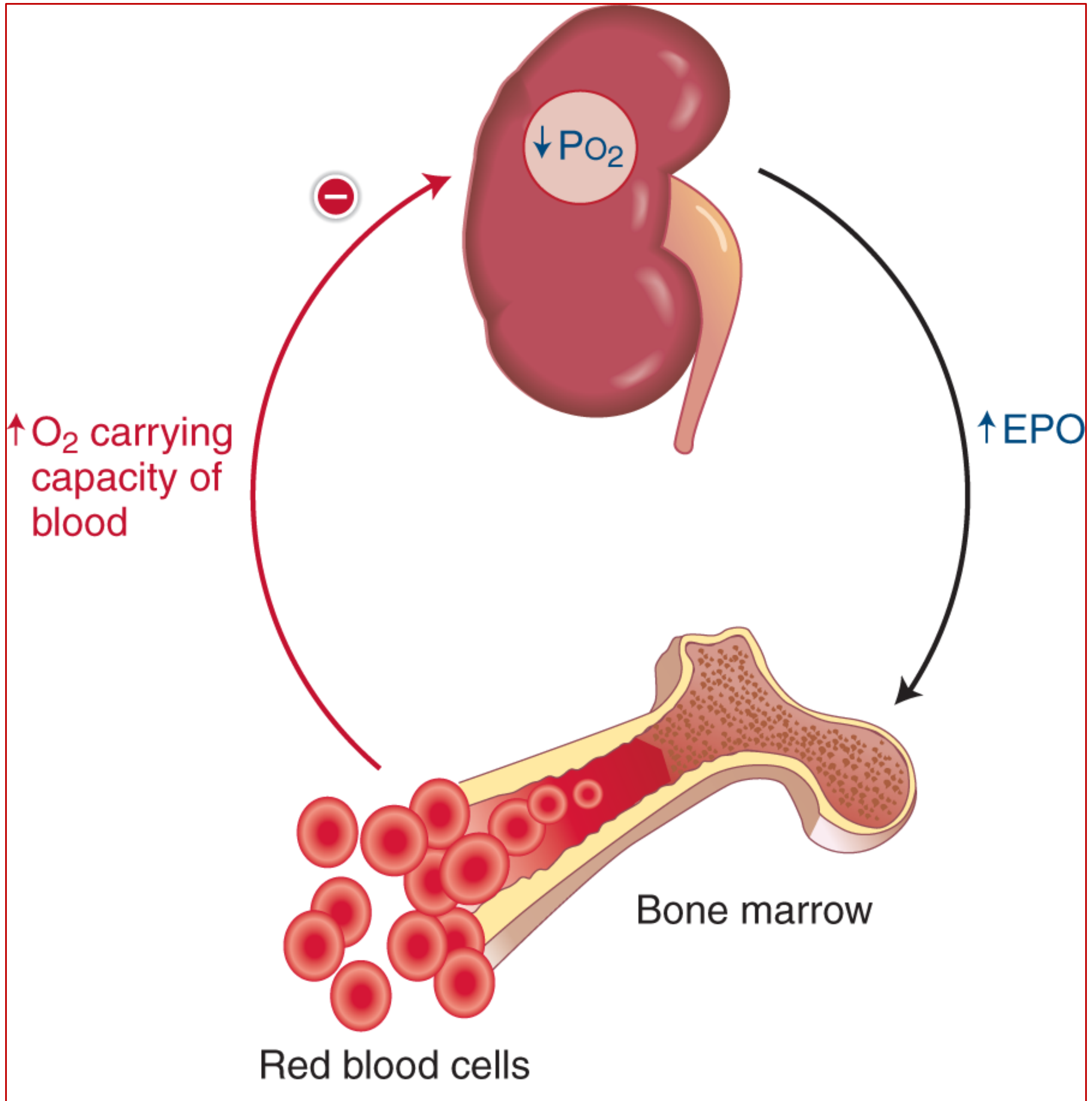
ESAs are recombinant analogs of EPO and mimic erythropoiesis

Physiology:

Erythropoietin (EPO) is expressed mainly by peritubular interstitial cells of the kidney in response to low oxygen levels (hypoxia or anemia).

Pharmacology: Like natural EPO…

- ESAs bind the erythropoietin receptor (EPOR) on erythroid progenitor cells in bone marrow.
- The receptors dimerize and activate the JAK/STAT signaling pathway → PI3/AKT and MAP
- Increase survival and proliferation of erythroid precursors:
  - Reduced apoptosis of progenitor cells
  - Increased maturation into reticulocytes
  - Increased RBC production



The Big Picture: Medical Biochemistry > The Urinary System Figure 18-9

ESAs: Indications and Monitoring

Goal of ESA therapy: To reduce the need for RBC transfusions by increasing Hb to the lowest level that alleviates anemia while minimizing thrombotic and cardiovascular risk.

	Epoetin alfa / Darbepoetin alfa	
Treatment indications	• Anemia of CKD • Symptomatic myelodysplastic syndrome • Chemotherapy-induced anemia in selected cancer patients.	• Chronic kidney disease is the most common reason for ESA therapy. • ESAs have been shown to increase serious risks, especially thromboembolism and worse cancer outcomes, without improving survival
Monitoring	• Hemoglobin: once weekly until maintenance dose established, then monthly • Iron stores (TIBC, ferritin) prior to and during therapy (ensure iron intake) • Blood pressure • For seizures for first few months due to increased blood viscosity and rapid changes in blood pressure – seizures are uncommon. • CKD patients: Serum chemistry and fluid balance; neurologic sx (seizures)	

ESAs: Adverse Effects

ESAs increase blood viscosity, thrombotic risk, and can worsen hypertension.

US Boxed Warnings: ESAs increase the risk of death, MI, stroke, venous thromboembolism (VTE), thrombosis of vascular access

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- Hypertension
- Cardiovascular events:
  venous thromboembolism,
  myocardial infarction, stroke
- Dermatologic effects reported:
  Rash,
  Stevens-Johnson syndrome / toxic epidermal necrolysis

Chronic kidney disease (CKD): ***At Hb levels >11***, CKD patients are at greater risk of cardiovascular adverse events.

- Pure red cell aplasia due to development of neutralizing anti-EPO antibodies is reported primarily in CKD patients receiving epoetin alfa SubQ.

Chemo-induced anemia:
VTE, Shortened overall survival and reduced time to tumor progression or recurrence

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CONTRAINDICATIONS:
Patients with uncontrolled hypertension, severe red cell aplasia that begins after treatment, history of hypersensitivity reactions, combination therapy with other ESAs

Check your knowledge

What is the main source of erythropoietin in adults?

Why is the bioavailability of subcutaneous epoetin-alfa so low?

How do ESAs stimulate erythropoiesis?

What is the effect of activating the signaling pathway?

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Check your knowledge

What is the primary clinical goal of ESA therapy?

In which clinical condition is ESA therapy most frequently used?

What laboratory parameter is used to guide EPA dose adjustments?

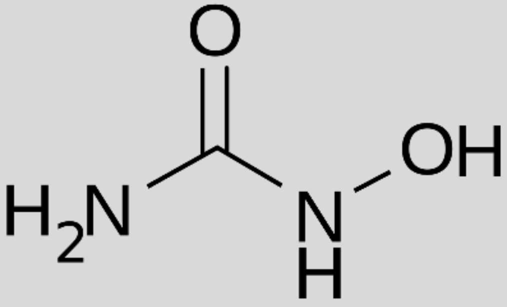
What is the major mechanism-related adverse event of ESA therapy?

Disease-modifying therapies for sickle cell disease

✳Hydroxyurea – first-line therapy

Crizanlizumab

L-Glutamine (pharmaceutical grade)



These medications are intended to be started when the individual is well.

They are **not appropriate** for treating acute pain.

Pharmacokinetics: Hydroxyurea

A	Oral; well absorbed
D	Protein binding ~80% Widely distributed including the brain. Concentrates in leukocytes and erythrocytes
M	Hepatic (urease by intestinal bacteria) Bioavailability 100%
E	Urine ~40% of the dose in patients with sickle cell anemia

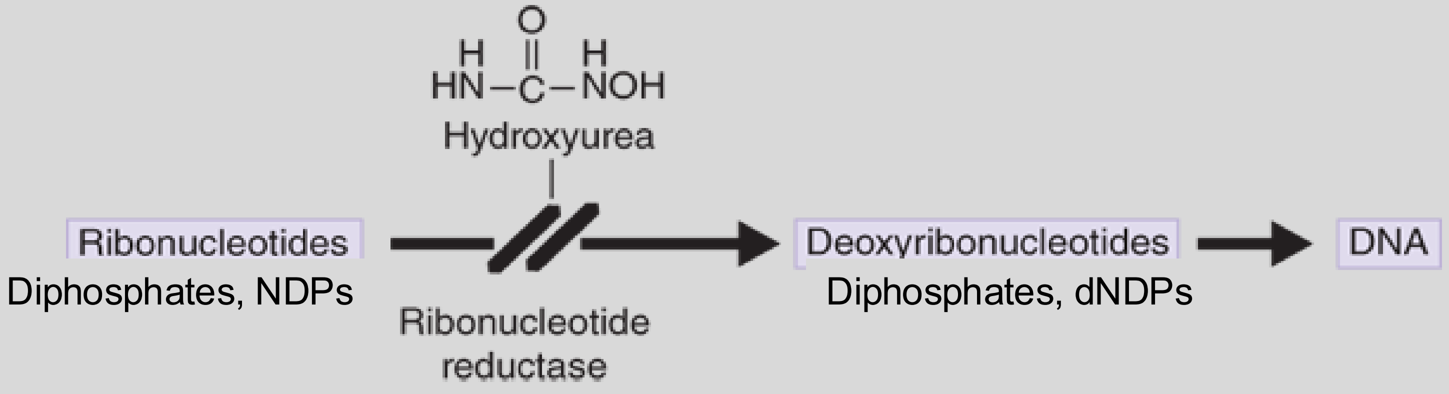
Effect **HbF induction: Fetal hemoglobin increases in 4-12 weeks.**

Sickle Cell Anemia Goal of Therapy and Mechanisms

Goal	To reduce sickling by increasing fetal hemoglobin(HbF) and improving red cell physiology. Increasing HbF inhibits hemoglobin S polymerization, which reduces vaso-occlusion, hemolysis, and disease complications.
Hydroxyurea	RNR inhibitor: Inhibits ribonuclease reduction and DNA synthesis, which halts the cell cycle in S phase, induces stress erythropoiesis and enhances γ-globin gene transcription $\rightarrow$ increases levels of HbF. See next slide.
Crizanlizumab	P-selectin inhibitor: Prevents adhesion of P-selectin to the endothelium, which can help manage pain crises P-selectin recruits leukocytes to the site of occlusion and facilitates interactions between platelets and leukocytes, which exacerbates inflammation and contributes to the severity of pain crisis.
L-Glutamine	Oxidative stress reduction: Enhances NADH/NAD+ balance in red blood cells, which supports glycolysis for ATP generation, maintains hemoglobin iron in the Fe2+ (reduced) state, and improves membrane integrity (a downstream effect).

Hydroxyurea Mechanism in Sickle Cell Anemia

Hydroxyurea binds iron molecules in catalytic subunit of RNR and inhibits the enzyme, halting cell cycle in the S phase.



Biochemistry, DNA biosynthesis:

NDPs – RNR → dNDPs → DNA synthesis

RNR catalyzes the reductive conversion of ribonucleotides to deoxyribonucleotides (rate limiting step).

Stress erythropoiesis:

Hydroxyurea binds iron molecules in catalytic subunit of RNR→ slows DNA synthesis in rapidly dividing RBCs→ mild bone marrow suppression.

↓

The bone marrow shifts toward production of new immature RBCs – stress erythroid precursors – that preferentially express γ-globin genes→ ↑HbF production

Supporting mechanism:

Hydroxyurea is converted to nitric oxide (NO) → stimulates soluble guanylyl cyclase → ↑cGMP

cGMP signaling enhances γ-globin gene transcription →↑ HbF

Hydroxyurea's contributing actions reduce vaso-occlusion:

HU-induced mild bone marrow suppression lowers neutrophils, platelets, and adhesion molecules → decreases inflammatory cell-adhesion molecule interactions and microvascular plugging

→ Vaso-occlusive pain crises are reduced.

Multiple beneficial effects of hydroxyurea for sickle cell anemia.

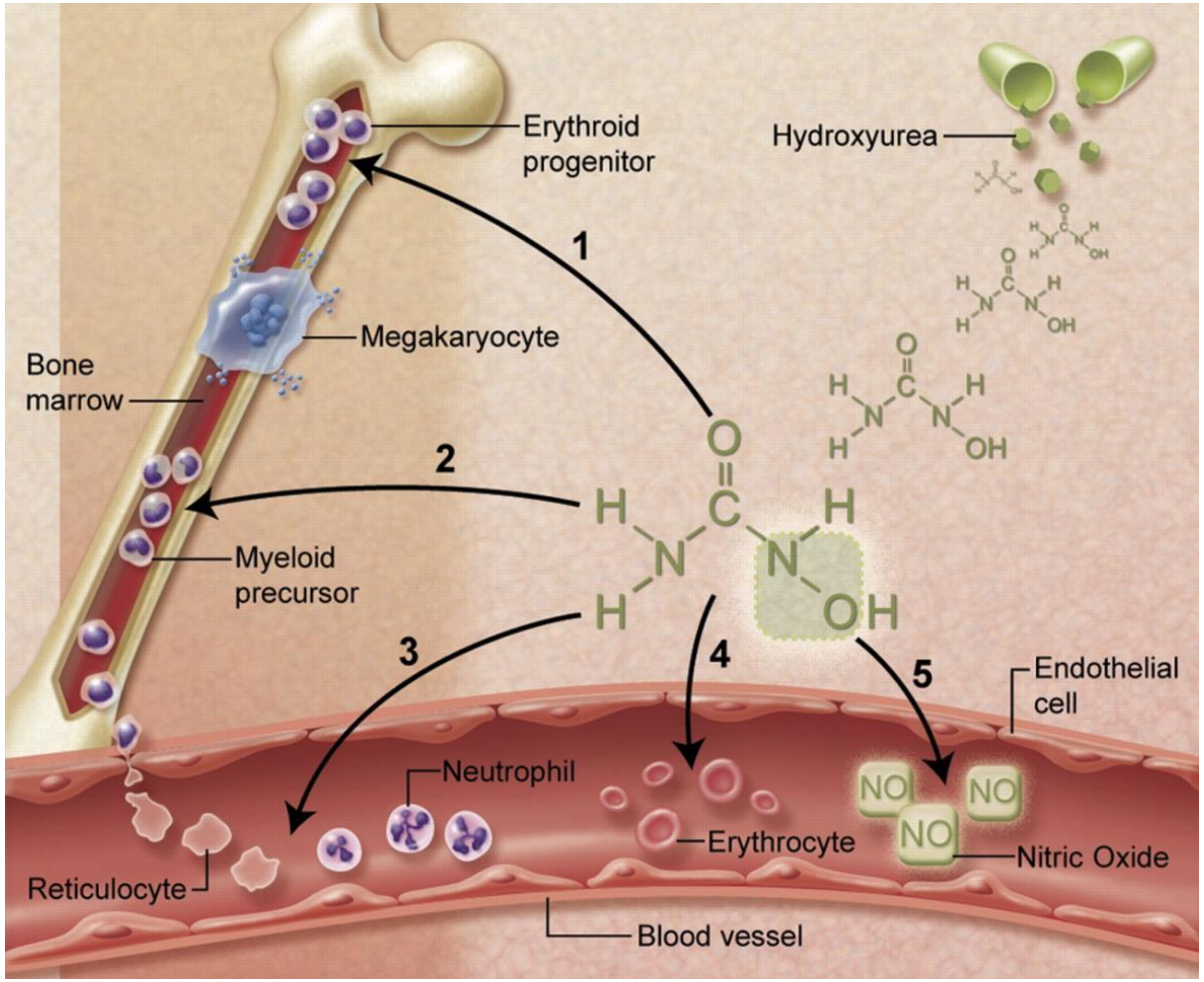
(1) Fetal hemoglobin induction by nitric oxide-dependent activation of soluble guanylyl cyclase, which alters kinetics of HbS polymerization;

(2) Lower neutrophil and reticulocyte counts from ribonucleotide reductase inhibition and marrow cytotoxicity;

(3) Decreased adhesiveness and improved rheology of circulating neutrophils and reticulocytes;

(4) Reduced hemolysis through improved erythrocyte hydration, macrocytosis, and reduced intracellular sickling;

(5) Nitric oxide (NO) release with potential local vasodilatation and improved vascular response.



Russell E. Ware Blood 2010;115:5300-5311

Hydroxyurea Adverse Effects

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Hydroxyurea is safe for long-term use in sickle cell disease, when appropriately monitored.

- **Myelosuppression:**
  neutropenia (mainly), anemia, thrombocytopenia
- GI distrubances
- Dermatologic reactions (mild)
- Hyperpigmentation → darkening of skin and nails
- Hair loss

Frequent monitoring of CBC with differential, especially during drug initiation and dose titration.

Reproductive Implications

- Teratogenic in animal reproduction studies
- May cause fetal harm
- May damage spermatozoa and testicular tissue; azoospermia or oligospermia
- Males and females of reproductive potential should use effective contraception during and for at least 6 months to 1 year following therapy.

Breast Feeding Not recommended

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Check your knowledge

What clinical complications of sickle cell disease are reduced with hydroxyurea therapy?

What is the goal of hydroxyurea therapy in sickle cell anemia?

How does the effect of hydroxyurea improve sickle cell disease?

What is the principal mechanisms of hydroxyurea in sickle cell anemia?

Check your knowledge

What are secondary mechanisms?

What is the major dose-limiting toxicity of hydroxyurea?

What cosmetic effects can occur with long-term therapy?

What laboratory test must be monitored during therapy?

Is long-term hydroxyurea therapy in sickle cells disease safe?

Disease-modifying therapies for sickle cell disease in brief

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Therapy	Hydroxyurea	Crizanlizumab	L-glutamine
Age	All, including infants	16 years and older	5 years and older
Route	Oral once daily	IV every 4 weeks after initial doses on weeks 0,2,and 4	Oral twice daily
MOA	↑ HbF levels, RBC water content, deformability of sickled cells, and alters adhesion of RBCs to endothelium	Anti-P-selectin: reduces interactions between endothelial cells and circulating RBCs	Alters redox state of RBCs: L-Glutamine is a NAD+ precursor. NAD+ is low in sickled RBCs. L-Glutamine can reduce oxidative stress and improve deformability of RBCs
Monitoring	Frequent monitoring of CBC and dose titration especially during drug initiation	None	None
Benefits	• Reduced vaso-occlusive complications and acute chest syndrome • ↓ transfusion requirement • Long-term safety/efficacy data	• Reduced frequency of acute pain episodes • Can be combined with hydroxyurea • Good safety profile	• Reduced frequency of acute pain episodes • Can be combined with hydroxyurea • Good safety profile
Comments	• Neutropenia, hyperpigmentation, hair loss • Caution prior to conception (males/females) and during first trimester of pregnancy	IV access, hospital admin. Arthraligias, back pain Infusion reactions possible	Use pharmaceutical grade powder pkts Mild GI symptoms Long-term efficacy data are lacking

Summary: Pharmacology of Anemias

- Any treatable cause of anemia should be corrected.

Iron deficiency anemia:

- Elemental iron supplementation supports hemoglobin synthesis / erythropoiesis by restoring iron stores (ferritin) and circulation iron (transferrin) saturation. IV iron is given for chronic anemias that cannot be maintained with oral iron alone.
- Oral iron supplements are best absorbed on an empty stomach. GI discomfort, constipation, and dark stools are common. IV iron products may cause injection reactions, hypersensitivity reactions, and exacerbate asthma and inflammatory arthritis.
- Reticulocyte count rises in ~1 week; Hb rises in ~2-3 weeks with iron supplement.

Megaloblastic anemia

- Folate (vitamin B9) supplementation replenishes tetrahydrofolate needed for DNA synthesis and corrects thymidylate production in rapidly dividing cells. It corrects anemia in days to weeks. It does not correct neurological deficits resulting from vitamin B12 deficiency.
- Vitamin B12 restores the cofactor for methylmalonyl-CoA mutase and methionine synthase, supports DNA synthesis and myelin integrity. Vitamin B12 (cyanocobalamin) administration is used in the treatment of pernicious anemia, malabsorption (Crohn disease, gastric bypass), strict vegetarian diet. Neurologic symptoms improve but may be irreversible if deficiency is long-standing. IM or oral are effective depending on the etiology.
- Hydroxocobalamin is used for B12 deficiency and as an antidote for cyanide poisoning.

Erythropoiesis-stimulating agents (EPAs):

- Recombinant erythropoietin that directly bind EPORs on erythroid precursors, activate JAK-STAT pathway, resulting in RBC production.
- They are used in chronic kidney disease (primary use) and selected cancer patients receiving myelosuppressive chemotherapy. The goal of therapy is to reduce the need for transfusion using the lowest effective dose. (The goal is not to normalize hemoglobin.)
- Risks: hypertension and thrombotic events – eg, MI, stroke, venous thromboembolism – and may shorten the overall survival or time to tumor progression or recurrence in cancer patients.

Sickle cell anemia

- Hydroxyurea is and RNR inhibitor that increases hemoglobin F production by inducing γ-globin gene expression. It also increases nitric oxide, improving vasodilation.
- HbF inhibits HbS polymerization, which reduces sickling, veno-occlusive crisis, and acute chest syndrome.
- The dose limiting toxicity is myelosuppression, mainly neutropenia. Monitoring CBC with differential is required.
- Hydroxyurea has long-term safety and efficacy data in the treatment of sickle cell anemia.
- Crizanlizumab and L-glutamine are options and may be used as adjuncts to hydroxyurea therapy.

Check your knowledge

What are the goals of iron supplementation therapy?

- To fully replace iron balance, identify and treat the underlying cause, and prevent recurrence

What is the first-line treatment of iron deficiency anemia?

- Oral iron supplement

What laboratory value best reflects total body iron stores?

- Serum ferritin directly measures iron stores. Low level is the most sensitive and specific test.
- Other critical markers: low hemoglobin/hemotocrit, low MCV, low serum iron, high total iron binding capacity.

Which form of iron is absorbed by enterocytes.

- Fe2+: Heme iron (animal foods) or Fe3+ (plants) → DCytB → Fe2+

Why should oral iron supplements be taken on an empty stomach?

- Polyphenols in coffee and tea, calcium-rich foods and fiber form insoluble complexes with non-heme iron, which reduces iron absorption.

Check your knowledge

How does the body regulate the amount of iron that is absorbed?

- The body's needs: Less iron is absorbed when iron stores are adequate. More iron is absorbed when iron stores are low.
- A lower percentage of iron is absorbed with higher doses (saturable absorption).

What are some important drug interactions?

- Iron reduces absorption of levothyroxine, quinolones, levodopa and others
- Iron absorption is reduced by antacids, bisphosphonates, tetracycline

When is IV iron infusion preferred?

- Rapid iron correction is needed / Oral iron is not tolerated or effective / Impaired absorption

What vitamin enhances oral absorption of iron?

- Vitamin C a strong reducing agent: $Fe^{3+} \rightarrow Fe^{2+}$

Why is gastric acid important for absorption of iron from plant-based food?

- Acid releases Fe^{3+} from proteins in food, solubilizing it so it is available for conversion to Fe^{2+}.

Check your knowledge: Folate and Vitamin B12 supplements

What vitamin deficiencies lead to megaloblastic anemia?

- Folate and vitamin B12.

How do the mechanisms of anemia by the two vitamins differ?

- Folic acid deficiency leads to diminished tetrahydrofolate (THF). Therefore, THF is not available for the one-carbon transfers needed in the synthesis of thymidylate and purines essential for DNA replication during cell division, stalling cell division in the erythroid precursors.
- B12 deficiency traps folate as unusable methylfolate, which stops production of the thymidylate and purines needed for RBC synthesis – functional folate deficiency.

Which vitamin deficiency leads to neurological symptoms and why?

- B12 deficiency causes dysfunction of the methylmalonyl-CoA mutase.
- Methylmalonyl-CoA and propionyl-CoA accumulate resulting in production of abnormal fatty acids that are incorporated into neuronal lipids and disrupt the structure and function of the myelin sheath, resulting in neurological symptoms:
- Paresthesia in peripheral nerves and weakness; progresses to spasticity, ataxia, neuropsychiatric changes (may be irreversible).

Check your knowledge: Folate and Vitamin B12 supplements

How does absorption of folic acid and vitamin B12 differ?

Folic acid → DHFR in duodenum / proximal jejunum → liver

Vitamin B12 → complex with intrinsic factor → absorbed in terminal ileum → liver

Details:

- Folic acid is present in foods as polyglutamates, which are hydrolyzed to monoglutamate forms by brush-border glutamate carboxypeptidase.
- Folic acid (including supplemental folic acid) is reduced by dihydrofolate reductase (DHFR) in the mucosal epithelial cells of the duodenum and proximal jejunum to tetrahydrofolate, subsequently methylated in the liver, secreted into bile, and undergoes enterohepatic circulation (the liver is major regulator of folate). Folate is stored as polyglutamates in hepatocytes.
- In the duodenum, vitamin B12 binds intrinsic factor, which is produced by gastric parietal cells. The complex travels through the intestine to the distal ileum and binds specific receptors. The complex is internalized in ileal cells by endocytosis. B12 is separated from IF and released into the bloodstream.

Check your knowledge: ESA therapy

What is the main source of erythropoietin in adults?

- Specialized peritubular interstitial cells of the kidney in response to low oxygen levels (hypoxia or anemia).

Why is the bioavailability of subcutaneous epoetin-alfa so low?

- ESAs are large molecules that slowly diffuse into lymphatics, which is capacity limited.
- Proteases, peptidases, macrophages in the interstitial space degrade the molecule

How do ESAs stimulate erythropoiesis?

- ESAs directly bind the erythropoietin receptor (EPOR) on erythroid progenitor cells in bone marrow.
- The receptors dimerize and activate the JAK/STAT signaling pathway → PI3/AKT and MAP.
- Note: ESAs do not stimulate release of EPO from.

What is the effect of activating the signaling pathway?

- Increased survival and proliferation of erythroid precursors:
  - Reduced apoptosis of progenitor cells
  - Increased maturation into reticulocytes
  - Increased RBC production

Check your knowledge: ESA therapy

What is the primary clinical goal of ESA therapy?

- To safely reduce the need for RBC transfusions by using the lowest dose needed to alleviate anemia.

In which clinical condition is ESA therapy most frequently used?

- Anemia of chronic kidney disease
- Cells in the renal cortex that product EPO are damaged or lost due to injury, inflammation, and fibrosis.

What laboratory parameter is used to guide EPA dose adjustments?

- Hemoglobin concentration

What is the major mechanism-related adverse event of ESA therapy?

- Increased risk of thrombosis that can lead to myocardial infarction and stroke.

Check your knowledge: Hydroxyurea therapy for sickle cell anemia

What clinical complications of sickle cell disease are reduced with hydroxyurea therapy?

- Vaso-occlusive pain crisis and acute chest syndrome.

What is the goal of hydroxyurea therapy in sickle cell anemia?

- To reduce sickling by increasing fetal hemoglobin (HbF) and improving red cell physiology.

How does the effect of hydroxyurea improve sickle cell disease?

- HbF reduces polymerization by HbS, which is the root cause of sickling and vaso-occlusive crisis.

What is the principal mechanisms of hydroxyurea in sickle cell anemia?

Stress erythropoiesis:

- Hydroxyurea binds iron molecules in catalytic subunit of RNR, which slows DNA synthesis in rapidly dividing RBCs, leading to mild bone marrow suppression.
- The bone marrow shifts toward production of new immature RBCs – stress erythroid precursors – that preferentially express γ-globin genes, thereby increasing ↑HbF production.

Check your knowledge: Hydroxyurea therapy for sickle cell anemia

What are secondary mechanisms?

- HU increases nitric oxide, which stimulates soluble guanylyl cyclase, increasing cGMP concentrations. cGMP signaling enhances γ-globin gene transcription $\rightarrow\uparrow$ HbF.
- HU-induced mild bone marrow suppression lowers neutrophils, platelets, and adhesion molecules, which decreases inflammatory cell-adhesion molecule interactions and microvascular plugging. Vaso-occlusive pain crises are reduced.

What is the major dose-limiting toxicity of hydroxyurea?

- Bone marrow suppression, especially neutropenia.

What cosmetic effects can occur with long-term therapy?

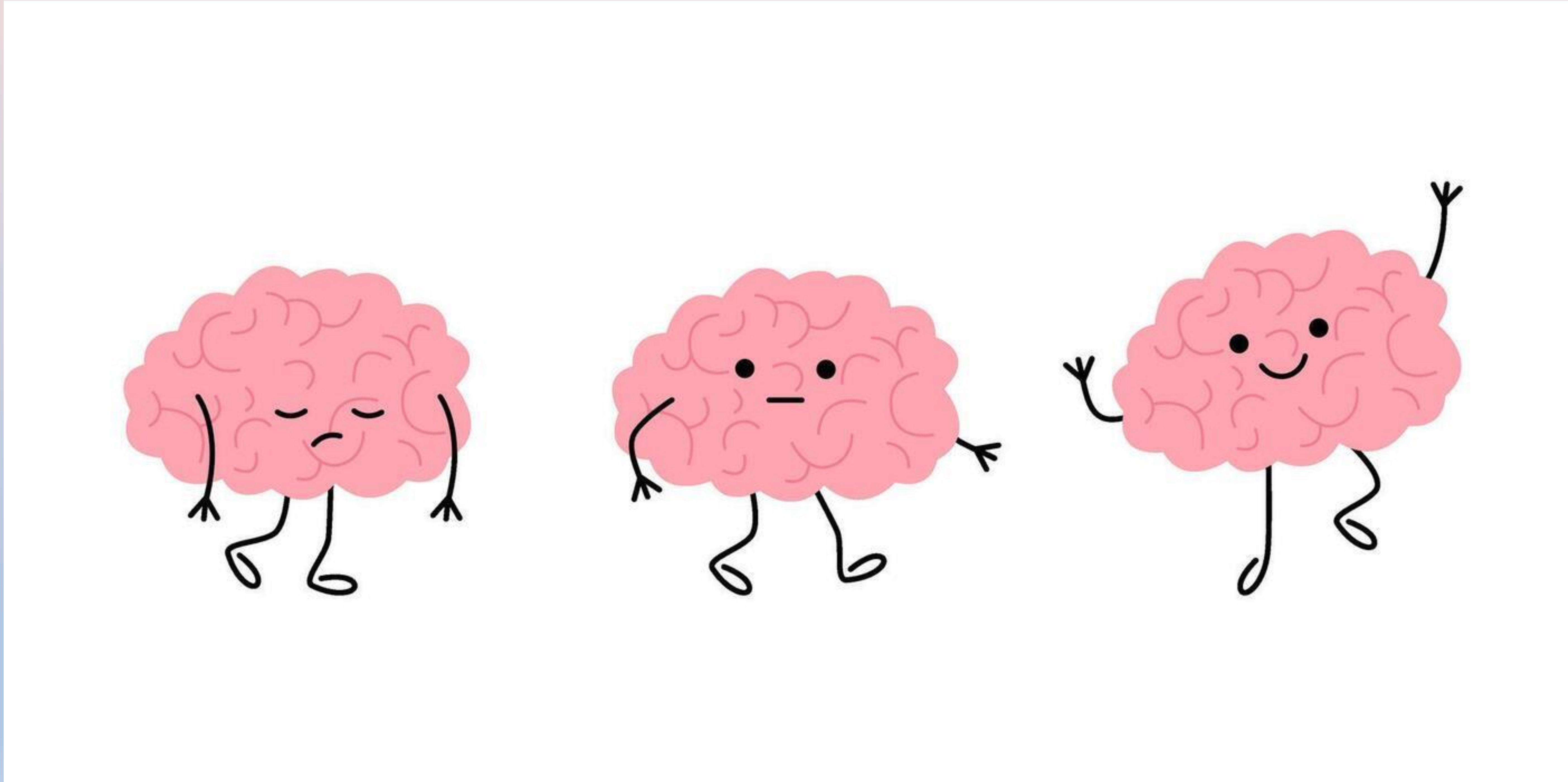
- Hyperpigmentation can cause darkening of skin and nails.

What laboratory test must be monitored during therapy?

- Complete blood count (CBC) with differential.

Is long-term hydroxyurea therapy in sickle cells disease safe?

- Yes, when appropriately monitored (based on long-term studies).



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Resources

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 33: Agents Used in Cytopenias; Hematopoietic Growth Factors

Access Medicine Goodman & Gilman's The Pharmacological Basis of Therapeutics, 14e, 2023; Chapter 45: Hematopoietic Agents: Growth Factors, Minerals, and Vitamins

Access Medicine Harrison's Principles of Internal Medicine, 21e, 2023; Chapter 99: Megaloblastic Anemias

Access Medicine Pathophysiology of Blood Disorders, 2e, 2017; Chapter 3: Overview of the Anemias

Access Medicine Williams Hematology, 10e, 2021; Chapter 38: Anemia of Chronic Disease; Chapter 50: Disorders of Hemoglobin Structure: Sickle Cell Anemia and Related Abnormalities

UpToDate Lexidrug monographs accessed March 2025

UpToDate: various topics; literature review current through Feb 2025

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